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Beyond Search Engine Optimization: How Large Language Models Are Redefining Surgeon Visibility

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Summary: Large language models (LLMs), such as ChatGPT, are rapidly transforming how patients identify and evaluate surgeons, marking the most significant shift in digital patient acquisition since the emergence of search engines. For decades, surgeon visibility online has depended on search engine optimization (SEO), a marketing strategy built around technical website performance, backlinks, and strategic content marketing designed to match keyword-based search behavior. This is in direct contrast to LLMs, which operate as "recommendation engines" and synthesize information across vast sources to generate personalized, conversational guidance in response to user queries. Rather than scanning ranked lists of links, patients increasingly can ask nuanced questions and receive narrative, context-sensitive answers. This shift fundamentally alters how expertise is recognized online. LLMs deemphasize traditional SEO signals and instead can emphasize more nuanced information ("language"), such as academic affiliation, peer-reviewed scholarship, institutional reputation, high-quality educational writing, and consistency across credible sources. This article outlines how LLMs form surgeon recommendations, why conventional SEO approaches are increasingly insufficient, and what practical steps surgeons can take to strengthen visibility in an artificial intelligence–mediated digital landscape. As generative artificial intelligence becomes embedded into everyday patient information-seeking, surgeons who adapt to this new recommendation paradigm can be best positioned for the next era of online discoverability. *(Plast Reconstr Surg Glob Open 2026;14:e7841; doi: 10.1097/GOX.0000000000007841; Published online 17 June 2026.)*

INTRODUCTION

For more than a decade, plastic and reconstructive surgeons have relied on search engine optimization (SEO), online reviews, and social media presence to establish visibility in an increasingly digital patient marketplace.[1,2] These strategies were built for a world in which patients discovered surgeons by typing keywords into a search box and scanning through pages of links. The emergence of large language models (LLMs), such as ChatGPT and other generative artificial intelligence (AI) systems, represents a fundamental shift in how patients and physicians seek and deliver medical information.[3–7]

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The term "large language models" encompasses a heterogeneous group of generative AI systems with varying architectures, training data sources, update frequencies, retrieval mechanisms, and safety alignment layers. These models are probabilistic generative systems trained on large corpora of text to predict the most likely next token in a sequence. They do not "search" databases in real time or independently verify credentials unless explicitly connected to external retrieval systems. Instead, responses reflect statistical associations learned during training. This architecture explains both their fluency and their limitations: when confident patterns exist in training data, outputs may seem authoritative; when gaps exist, models may generate plausible but inaccurate statements, a phenomenon commonly referred to as hallucination. Some deployed systems incorporate retrieval-augmented generation, in which external documents are programmatically retrieved and incorporated into the model's response to improve factual grounding. However, not all consumer-facing LLM interfaces rely consistently on retrieval pipelines, and variability across platforms may influence output reliability. Accordingly, behaviors described in this article reflect general patterns

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observed across contemporary conversational generative AI systems rather than uniform functionality across all platforms.

Although traditional search engines remain the primary pathway for local patient acquisition, LLMs are increasingly shaping early-stage medical information-seeking and may represent the most significant structural shift in digital discoverability since the emergence of search engines. Specifically, rather than "searching," patients can now "ask," posing conversational, highly contextual questions, and receive a concise, situationally specific answer generated from thousands of sources in seconds. In a departure from traditional SEO, these responses are not governed by backlinks, keyword density, or paid placement. Instead, they reflect patterns derived from a variety of sources, including medical literature, institutional reputations, clinician-authored materials, and professional consensus.

This shift from "search" engines to "recommendation" engines has profound implications for surgeons. Surgeons who previously relied on SEO-driven tactics may lose ground as LLMs emphasize entirely different signals of credibility. As LLM systems increasingly mediate early phases of patient decision-making, understanding how they form recommendations and how surgeons can authentically position themselves within this new online ecosystem has become essential. Importantly, the term "recommendation" refers to the surfacing of names within conversational outputs rather than a formal recommendation mechanism. Herein, we provide a conceptual commentary synthesizing current literature, publicly documented model characteristics, and observed patterns in LLM outputs, rather than a formal experimental evaluation of model performance.

FROM SEARCH TO DIALOGUE: HOW PATIENTS NOW SEEK SURGEONS

For decades, digital patient acquisition was shaped by a predictable pathway: entering keywords into a search engine and comparing the list of local surgeons that is provided. Search engine ranking depended on factors such as website age, backlink structure, keyword architecture, mobile performance, and paid advertisements. Technically optimizing this process to be favored in the search is formally known as SEO.

Today, this pathway is expanding. Although patients continue to use traditional search engines for location-based queries, many now supplement those searches with generative AI tools for medical queries, a trend accelerated by their easy integration into smartphones, computer operating systems, and search platforms. Instead of web searching "best breast surgeon near me," patients may now ask coherent questions, such as "who is known for natural-appearing breast augmentation?" or "which surgeons in New York City specialize in preservation breast augmentation?" and receive a coherent narrative answer in return (rather than a list of website links). For surgeons who depend on digital patient acquisition, learning to optimize this process is essential.

Takeaways

Question: How does the rise of large language models change the way patients discover and evaluate plastic surgeons online, and why do traditional search engine optimization strategies no longer reliably determine surgeon visibility?

Findings: This narrative review analyzes how large language models function as recommendation engines rather than search engines and contrasts their behavior with traditional search engine optimization–based discovery.

Meaning: As patients shift from searching to asking, surgeons must invest in authentic expertise, academic credibility, and high-quality educational content to remain visible in an artificial intelligence–mediated healthcare marketplace.

HOW LLM RECOMMENDATIONS DIFFER FROM SEARCH ENGINE RANKINGS

LLM recommendations differ from SEO in numerous ways. Specifically, LLMs do not use backlinks, domain authority, keywords, or paid promotions, which are all mechanical inputs that form the foundation of SEO. Unlike traditional search engines, LLM outputs are not primarily driven by live backlink indexing or bid-based ranking systems. Rather, they rely on model training data, statistical associations, contextual understanding, and reputation signals embedded in language, rather than metadata.[8] However, technical website infrastructure, such as structured schema markup, consistent credential data, and machine-readable formatting, continues to influence how information is ingested and represented across digital ecosystems. Thus, a surgeon with technically optimized SEO but minimal academic or professional footprint may appear highly visible in a traditional search engine but be absent in LLM recommendations.

When recommending surgeons, LLMs also uniquely integrate information across domains. They can synthesize a variety of sources, including traditional academic endeavors (published scientific literature, professional society roles and contributions, book chapters) and alternative media (lay news articles, hospital affiliations, educational blogs or videos, consistent positive mention across diverse sources).[9] Notably, this multidomain integration is something SEO cannot replicate.

LLMs are also adept at contextual personalization that evolves through interaction. This too is different from traditional search engines, as LLMs can adapt to user phrasing, prior questions, stated preferences, perceived intent, and conversational nuance. For example, a user who has a history of asking for conservative aesthetic modifications can receive a different recommendation for a breast augmentation surgeon than someone asking for dramatic transformations (ie, surgeon recommendations can shift dynamically depending on how the patient engages; it is not a fixed list). This personalization makes LLM outputs feel more relevant to the patient but is harder to predict or influence for the surgeon.

Importantly, LLMs do not presently allow sponsorship or bid-based ranking. Advertisements may eventually appear in some interfaces, but foundational model behavior is not currently influenced by user payment. This removes one of the core SEO mechanisms through which surgical practices historically increased visibility.

WHAT SIGNALS LLMS ACTUALLY USE TO RECOMMEND SURGEONS

Although proprietary, LLM behavior is consistent enough to identify major categories of signals that influence recommendations. One major category that influences recommendations is academic and institutional credibility. LLMs' responses frequently reflect associations between recognized academic institutions and perceived expertise, as well as publicized leadership roles in national organizations and publications in peer-reviewed journals. Although proprietary model weighting is not publicly available, this pattern may be consistent with training on large corpora that include academic publications, institutional biographies, and peer-reviewed literature.[10,11] Though not fully accurate, these serve as high-confidence proxies for expertise in the model's reasoning. In practice, surgeons with substantial academic presence and repeated mention in scholarly contexts may be more likely to appear in generated responses, although this relationship has not yet been formally quantified.

LLMs also reflect associations between textual quality, not just keyword volume. Signals include high-quality educational patient resources, precise terminology, coherent writing, and alignment with consensus medical language. Further, models can seek correlation of reputational consistency across data sources, and a surgeon mentioned on institutional pages, in news interviews, in peer-reviewed literature, and in patient testimonials is more likely to be recommended. Thus, surgeons with formal websites that present information such as patient-focused textbooks are more likely to be surfaced, as LLMs can detect tone, expertise, and narrative coherence.

Importantly, LLMs do not directly evaluate or "verify" expertise. Instead, they generate responses based on statistical patterns learned during training across extensive text corpora. A surgeon whose name, credentials, and institutional affiliation frequently co-occur with specialty-specific terminology in academic publications, institutional biographies, media interviews, and educational materials is more likely to be probabilistically associated with that domain of expertise in model outputs. In contrast to search engines, which rank pages based on live backlink networks and keyword architecture, LLM outputs reflect learned language associations across distributed sources. As a result, consistency of professional identity across multiple credible contexts may increase the likelihood of being surfaced in conversational responses. Practically, this suggests that expertise becomes visible not merely through technical optimization, but through repeated, coherent representation of one's specialty across academic literature, institutional pages, structured website content, and authoritative educational writing.

LIMITATIONS OF SEO-ONLY STRATEGIES IN AN AI-INTEGRATED ECOSYSTEM

SEO worked because search engines (especially Google) rewarded technical compliance such as fast websites, keywords, backlinks, and structured data, and SEO remains foundational to digital visibility. However, LLMs do not value these technical aspects in the same way, and LLM-mediated recommendations seem less sensitive to purely mechanical optimization strategies (eg, keyword saturation or backlink volume) and more aligned with substantive signals of expertise, clarity, and cross-platform reputational consistency. Thus, SEO-optimized content, such as "Top 10 Reasons" lists, may be deprioritized because LLMs detect template phrasing, promotional tone, and low informational density.

PRACTICAL GUIDANCE: HOW AESTHETIC SURGEONS CAN IMPROVE LLM VISIBILITY

Technical Infrastructure (Still Essential)

It is still important to build a strong, broad digital footprint that clearly states training, specialties, and unique value, as well as maintain accurate technical schema markup on the website.[12] These include MedicalOrganization schema, Physician schema, and LocalBusiness schema, as well as accurate and updated credentials, specialties, consistent NAP (name, address, phone) data, verified institutional pages, and structured review and photograph data in crawlable formatting.[13] These are vital technical implementations that improve how LLMs "understand" an online identity and choose to provide recommendations.

Content Authority (Increasingly Weighted)

An increasingly important facet of online presence is the publication of authoritative content and niche expertise, as LLMs seem to favor professionals who are recognized as subject-matter experts. This can be optimized by publishing peer-reviewed research in the relevant subject area, posting short patient-facing technical writing on the website, and contributing expert quotes to reputable media. Further, participating in academic or professional collaborations, such as journal editorial boards or professional society leadership, also creates authoritative signals, improving a surgeon's appearance as a subject-matter expert. Ultimately, anything that becomes officially indexed online and can be referenced improves online visibility.

STRUCTURAL BIASES AND LIMITATIONS OF LLM-GENERATED SURGEON RECOMMENDATIONS

There are plenty of new challenges introduced by LLMs. Models have been known to invent credentials and publications, which pose reputational and ethical risks.[14] Additionally, their recommendations are proxies for credibility, not validated measures of outcome or skill. Beyond geographic concentration and academic emphasis, LLM-mediated visibility may intersect with long-standing structural inequities in medicine. Because LLMs learn patterns from existing text corpora, they reflect historical distributions of authorship, institutional prominence, and media representation. Surgeons affiliated with large academic centers are more likely to have extensive publication histories, institutional biographies, and media citations, which are signals that may increase probabilistic association with expertise in generated outputs.

This dynamic may disadvantage community-based practitioners who provide high-quality care but have a limited scholarly footprint. Similarly, early-career surgeons may have fewer publications and media references, reducing their textual representation in training data. Gender disparities in academic authorship and leadership representation may also translate into differential digital visibility. In addition, non–English-speaking physicians or surgeons practicing in lower resource or non–English-dominant regions may be underrepresented if training corpora are disproportionately English-language and Western-centric. As a result, LLM-generated recommendations risk reinforcing existing professional hierarchies rather than purely reflecting clinical quality. Ongoing evaluation, transparency, and inclusive data representation will be critical to ensure that AI-mediated discovery does not inadvertently perpetuate inequities in healthcare access and professional recognition.

Finally, and importantly, this article does not present structured experimental testing of LLM recommendation outputs. Formal studies using standardized prompt design, multimodel comparison, and quantitative assessment of recommendation frequency and reproducibility, as well as formal empirical validation of affiliation weighting or publication-associated surfacing, are needed to validate and refine these observations.

PATIENT SAFETY CONSIDERATIONS

Beyond structural bias, LLM-mediated surgeon recommendations introduce important patient safety considerations as the regulatory landscape for medical AI continues to evolve. In the United States, the Food and Drug Administration has issued guidance on clinical decision support software and risk-based oversight of AI-enabled medical devices. In the European Union, the AI Act introduces a tiered risk classification system, with healthcare applications often falling into high-risk categories requiring transparency, documentation, and human oversight. Although consumer-facing LLMs used for informational queries may not currently be regulated as medical devices, their influence on patient decision-making raises important questions about accountability, transparency, and professional responsibility.

First, generative models do not independently verify licensure status, board certification, complication history, or scope of practice. Outputs are generated based on language patterns rather than real-time credential validation. As a result, recommendations may omit critical contextual factors such as procedural specialization, geographic feasibility, or individual patient risk profiles.

Second, aesthetic surgery exists within a uniquely marketing-driven digital environment. If promotional or commercially oriented content is disproportionately represented in publicly available text corpora, model outputs may inadvertently reflect that emphasis. Although LLMs are not designed to promote specific individuals, they may reproduce patterns embedded in training data that blend educational and promotional language.

Third, younger and digitally native patients may attribute high epistemic authority to conversational AI systems. Narrative-style responses can feel personalized and definitive, potentially influencing decision-making before consultation with qualified professionals. These dynamics underscore the need for digital literacy, transparent model disclaimers, and reinforcement that AI-generated outputs are informational rather than prescriptive.

Ultimately, LLM systems should be viewed as informational intermediaries rather than clinical gatekeepers. Surgeons, institutions, and policymakers must ensure that AI-mediated discoverability does not replace credential verification, shared decision-making, or regulatory oversight mechanisms designed to protect patient safety.

CONCLUSIONS

The digital ecosystem governing surgeon discoverability is undergoing its most significant transformation since the birth of search engines. As conversational AI becomes increasingly integrated into digital ecosystems, patient information-seeking behaviors are evolving beyond purely keyword-based search. These systems emphasize entirely different signals than SEO: academic credibility, multisource reputation, and demonstrable expertise. For surgeons, adapting to this new reality requires introducing LLM-specific optimization strategies, such as substance-driven communication. Importantly, LLM-mediated visibility should be understood as an additive layer rather than a replacement for existing digital infrastructure. Local SEO, online reviews, Google Business Profiles, and geographically indexed searches remain central to patient acquisition. However, as conversational AI tools increasingly shape exploratory information-seeking, surgeons may benefit from integrating substance-driven authority signals alongside traditional optimization strategies. The future of digital discoverability is likely hybrid, not binary.

Nevertheless, the future is clear: surgeons who invest in authoritative, patient-centered educational materials and maintain a visible professional footprint are best positioned to be favorably represented in the LLM era. The practices that recognize and adapt to this shift through genuine expertise may be best positioned to define the next frontier of digital discoverability in aesthetic surgery.

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REFERENCES

1. Alessandri-Bonetti M, Liu HY, Palmesano M, et al. Online patient education in body contouring: a comparison between Google and ChatGPT. *J Plast Reconstr Aesthet Surg*. 2023;87:390–402.
2. Gould DJ, Grant Stevens W, Nazarian S. A primer on social media for plastic surgeons: what do I need to know about social media and how can it help my practice? *Aesthet Surg J*. 2017;37:614–619.
3. Bogdanovich B, Shah P, Patel PA, et al. Keeping up with AI evolution: ChatGPT-4o in surgery. *Am Surg*. 2025;91:5–6.
4. Boyd CJ, Hemal K, Sorenson TJ, et al. Artificial intelligence as a triage tool during the perioperative period: pilot study of accuracy and accessibility for clinical application. *Plast Reconstr Surg Glob Open*. 2024;12:e5580.
5. Kavian JA, Wilkey HL, Patel PA, et al. Harvesting the power of artificial intelligence for surgery: uses, implications, and ethical considerations. *Am Surg*. 2023;89:5102–5104.
6. Abi-Rafeh J, Bassiri-Tehrani B, Kazan R, et al. Preoperative patient guidance and education in aesthetic breast plastic surgery: a novel proposed application of artificial intelligence large language models. *Aesthet Surg J Open Forum*. 2024;6:ojae062.
7. Abi-Rafeh J, Xu HH, Kazan R, et al. Large language models and artificial intelligence: a primer for plastic surgeons on the demonstrated and potential applications, promises, and limitations of ChatGPT. *Aesthet Surg J*. 2024;44:329–343.
8. Maity S, Saikia MJ. Large language models in healthcare and medical applications: a review. *Bioengineering (Basel)*. 2025;12:631.
9. Wu K, Wu E, Wei K, et al. An automated framework for assessing how well LLMs cite relevant medical references. *Nat Commun*. 2025;16:3615.
10. Sim SZY, Chen T. Critique of impure reason: unveiling the reasoning behaviour of medical large language models. *Elife*. 2025;14:e106187.
11. Mansoor I, Abdullah M, Rizwan MD, et al. Reasoning with large language models in medicine: a systematic review of techniques, challenges and clinical integration. *Health Inf Sci Syst*. 2026;14:6.
12. van Berkel M. What is schema markup? A guide to structured data SEO. Available at https://www.schemaapp.com/schema-markup/what-is-schema-markup-a-guide-to-structured-data/. Accessed November 29, 2025.
13. Brickley D. Schema. Available at schema.org. Accessed November 29, 2025.
14. McGowan A, Gui Y, Dobbs M, et al. ChatGPT and Bard exhibit spontaneous citation fabrication during psychiatry literature search. *Psychiatry Res*. 2023;326:115334.